Comparing Grounded and Ungrounded LLM Response Accuracy on Domain-Specific Research Queries

Abstract

Large language models (LLMs) are increasingly deployed as research and decision-support tools in specialized domains such as law, medicine, and the sciences, where factual precision carries direct professional and safety consequences. This paper examines the empirical distinction between *ungrounded* LLMs, which answer solely from parametric memory acquired during pre-training, and *grounded* systems that couple generation with external retrieval, most commonly through retrieval-augmented generation (RAG) [5]. Drawing on the hallucination literature [4], benchmark studies of factual precision [6, 7, 11], and domain-specific empirical audits in law [2, 8] and medicine [13], we synthesize evidence on how grounding affects response accuracy for domain-specific research queries. We find that grounding reliably reduces, but does not eliminate, factual error, and that the residual error surface shifts in character–from fabricated content to misattribution, misapplication, and retrieval failure. We discuss current architectural approaches (Self-RAG, corrective RAG, agentic and iterative retrieval), articulate methodological and evaluation limitations, and identify research gaps around benchmark validity, retrieval-generation interaction effects, and domain transfer. We conclude that grounding is a necessary but insufficient condition for reliable domain-specific accuracy, and that evaluation frameworks must move beyond binary hallucination counts toward provenance-aware, task-conditioned accuracy metrics.

Keywords: retrieval-augmented generation, grounded language models, hallucination, factual accuracy, domain-specific question answering, large language models, information retrieval, legal AI, clinical NLP

1 Introduction

Large language models (LLMs) such as GPT-4, PaLM, and their successors have demonstrated broad competence across natural language tasks, yet their reliability on domain-specific research queries–questions in law, medicine, science, and other technical fields that demand verifiable, source-attributable answers–remains contested. Because LLMs encode knowledge implicitly in billions of parameters learned from a fixed, dated training corpus, they are prone to *hallucination*: the generation of plausible-sounding but factually unsupported or fabricated content [4]. In domains where an incorrect citation, an invented case law reference, or a fabricated drug interaction could cause professional or physical harm, this failure mode is more than an academic curiosity.

Two broad architectural paradigms have emerged in response. *Ungrounded* (or "closed-book") LLMs answer purely from parametric memory, relying on knowledge distilled during pre-training and any subsequent fine-tuning. *Grounded* LLMs instead couple the generator with an external, queryable knowledge source–a document index, search engine, or knowledge base–at inference time, most prominently through retrieval-augmented generation (RAG) [5]. The grounding hypothesis holds that by conditioning generation on retrieved, verifiable evidence, a model can reduce reliance on potentially stale or absent parametric knowledge and instead "look up" the answer, improving factual accuracy and enabling provenance.

This paper undertakes a comparative synthesis of grounded versus ungrounded LLM response accuracy specifically for domain-specific research queries. We ask three questions. First, what does the empirical literature show about the magnitude of accuracy improvement, if any, that grounding provides over parametric-only generation, and how consistent is this improvement across domains? Second, what failure modes persist even in grounded systems, and how do they differ qualitatively from the failure modes of ungrounded generation? Third, what methodological and infrastructural gaps limit our ability to draw confident, generalizable conclusions about grounded-versus-ungrounded accuracy in professional research contexts? In addressing these questions we draw on hallucination surveys, factuality benchmarks, and–critically–field audits of deployed, real-world grounded systems in legal research [2, 8], which offer some of the only large-scale, preregistered evidence of grounded system performance under realistic domain-specific query conditions.

The remainder of this paper is organized as follows. Section 2 reviews background and related work on hallucination, retrieval augmentation, and domain-specific evaluation. Section 3 presents the main technical and thematic discussion, comparing accuracy patterns across grounded and ungrounded settings. Section 4 surveys current architectural approaches designed to strengthen grounding. Section 5 discusses research challenges and limitations. Section 6 identifies research gaps and future directions. Section 7 concludes.

2 Background and Related Work

2.1 Hallucination in Large Language Models

Ji et al. [4] provide a comprehensive taxonomy of hallucination in natural language generation, distinguishing *intrinsic* hallucinations (output that contradicts source content) from *extrinsic* hallucinations (output that introduces unverifiable information), and further separating *factuality* hallucinations from *faithfulness* hallucinations. This taxonomy is foundational to comparative accuracy research because ungrounded and grounded systems tend to fail in different regions of this taxonomy: ungrounded models are more prone to extrinsic factual fabrication, whereas grounded models shift risk toward faithfulness failures–generating content inconsistent with the retrieved evidence even when that evidence is present in context.

Complementary work has quantified hallucination through self-consistency and black-box detection methods. Manakul et al. [10] introduce SelfCheckGPT, a zero-resource technique that detects hallucinated content by sampling multiple stochastic generations from the same ungrounded model and measuring inter-sample consistency, on the premise that fabricated facts vary across samples while learned facts remain stable. Li et al. [6] construct HaluEval, a large-scale benchmark of both machine-generated and human-annotated hallucinated samples, showing that existing LLMs struggle to recognize hallucinations in their own or others' output even when explicitly prompted to do so. Lin et al. [7] propose TruthfulQA, showing that larger, purely parametric models can become *less* truthful on certain adversarially constructed questions because scale amplifies the model's tendency to imitate common human misconceptions present in training data–a finding directly relevant to the limits of scaling ungrounded systems for research-grade accuracy.

2.2 Retrieval-Augmented Generation and Grounding

Lewis et al. [5] introduced retrieval-augmented generation, combining a pretrained parametric sequence-to-sequence model with a non-parametric dense vector index (originally over Wikipedia), retrieved via maximum inner product search and marginalized over during generation. The original RAG paper reported state-of-the-art results on open-domain question answering and more specific, factual generation than purely parametric baselines. Subsequent work extended the grounding paradigm in several directions: Shuster et al. [12] showed that retrieval augmentation substantially reduces hallucination rates in open-domain conversational agents relative to matched parametric-only baselines, while Mallen et al. [9] demonstrated that the benefit of retrieval is highly non-uniform–parametric memory alone is often sufficient and even preferable for high-frequency, popular entities, while retrieval yields the largest gains for long-tail, rarely seen facts, precisely the profile of many specialized research queries.

Gao et al. [3] synthesize the rapidly diversifying RAG literature into a taxonomy of *naive*, *advanced*, and *modular* RAG architectures, cataloguing pre-retrieval query transformation, retrieval re-ranking, and post-retrieval compression techniques designed to improve the fidelity of the grounding evidence supplied to the generator. Asai et al. [1] propose Self-RAG, in which the model is trained to adaptively decide when retrieval is necessary and to emit explicit critique tokens evaluating the relevance and support of retrieved passages, addressing the failure mode in which naive RAG indiscriminately retrieves and incorporates irrelevant passages.

2.3 Domain-Specific Empirical Evaluation

Because most hallucination benchmarks (TruthfulQA, HaluEval, FActScore) evaluate open-domain or Wikipedia-adjacent knowledge, their generalizability to specialized professional domains is limited. Two literatures partially fill this gap. In medicine, Singhal et al. [13] introduce MultiMedQA and evaluate PaLM and its instruction-tuned variant Med-PaLM across medical licensing-style and consumer health questions, finding that accuracy scales strongly with model size and instruction tuning but that human clinician evaluation reveals persistent gaps in comprehension, reasoning, and potential-harm dimensions not captured by multiple-choice accuracy alone.

In law, Dahl et al. [2] present the first systematic empirical audit of legal hallucination, finding that general-purpose LLMs (ChatGPT 3.5, ChatGPT 4, Llama 2, PaLM 2) hallucinate on specific, verifiable questions about federal court cases between 58% and 88% of the time, and that models frequently fail to correct a user's false legal premises or to recognize their own hallucinations. Building directly on this finding, Magesh et al. [8] conduct a preregistered empirical evaluation of three commercially deployed, RAG-based legal research tools (Lexis+ AI, Westlaw AI-Assisted Research, and Ask Practical Law AI) that are explicitly marketed by their vendors as "hallucination-free." They find that, while grounding substantially reduces hallucination relative to general-purpose ungrounded chatbots such as GPT-4, the RAG-based commercial tools still hallucinate or produce unsupported responses between 17% and 33% of the time, and that reliability varies substantially across products and query types. This study is arguably the single most direct empirical comparison available between grounded, production-grade domain-specific systems and ungrounded general-purpose baselines answering comparable research queries, and it anchors much of the comparative analysis in Section 3.

Finally, Min et al. [11] propose FActScore, an atomic-fact-level factual precision metric, and use it to show that even a retrieval-augmented commercial system (PerplexityAI) substantially outperforms purely parametric instruction-tuned baselines on long-form biography generation, though it still fails to achieve full factual precision–again illustrating that grounding raises the accuracy floor without eliminating error.

3 Main Thematic and Technical Discussion

3.1 Two Architectures of Answering

An ungrounded LLM answering a domain-specific research query resolves the query entirely through a single forward pass over parameters shaped by pre-training and fine-tuning. Its "knowledge" is static as of a training cutoff, is distributed diffusely across weights rather than indexed by source, and offers no intrinsic mechanism for provenance: the model cannot, in general, distinguish a fact it memorized from a fact it confabulated, because both are produced by the same generative process [4]. A grounded LLM, by contrast, first issues or is issued a retrieval query against an external corpus (case law databases, PubMed, an enterprise document store, or the open web), receives a set of retrieved passages, and conditions generation on this evidence, typically via a prompt-concatenation or marginalization mechanism [5]. This architectural difference has three accuracy-relevant consequences for domain-specific queries.

Recency and long-tail coverage. Domain-specific research queries frequently concern recently decided cases, newly published trial results, or narrowly scoped technical facts that are underrepresented in pre-training corpora. Mallen et al. [9] show empirically that parametric-only QA accuracy degrades sharply as a function of entity popularity (a proxy for training-data frequency), while retrieval-augmented accuracy remains comparatively stable across the popularity distribution. Because professional research queries are disproportionately about long-tail entities–a specific statute subsection, a particular clinical trial, an obscure enzyme–this asymmetry favors grounded systems for exactly the query distribution most relevant to this paper's scope.

Fabrication versus misapplication. The legal hallucination audits offer the clearest side-by-side evidence. Dahl et al. [2] document that ungrounded general-purpose models frequently invent entire case citations, holdings, or procedural postures that do not exist. Magesh et al. [8] show that grounded, RAG-based legal tools rarely invent a citation wholesale–because the citation is typically drawn from retrieved case text–but instead more often *mischaracterize* what a real, retrieved case actually held, cite a real case for a proposition it does not support, or conflate the holding of one retrieved case with the facts of another. This is consistent with the intrinsic/extrinsic distinction of Ji et al. [4]: grounding suppresses extrinsic fabrication but does not guarantee intrinsic faithfulness to the retrieved evidence, and reported hallucination rates of 17–33% in production-grade grounded legal tools versus 58–88% in ungrounded general-purpose chatbots on comparable federal case queries represent a large but incomplete improvement–roughly a two-to-fourfold reduction in error rate rather than elimination.

Retrieval as a new failure surface. Grounding does not remove error; it relocates part of it upstream into the retrieval step. If the retriever fails to surface the relevant passage–because of vocabulary mismatch, sparse indexing of a niche sub-domain, or a query that is ambiguously phrased–the generator either fabricates in the absence of evidence (behaving, in effect, like an ungrounded model) or, if instructed to abstain, under-answers. Asai et al. [1] explicitly motivate Self-RAG by observing that naive RAG's fixed, indiscriminate retrieval frequently injects irrelevant passages that measurably *degrade* answer quality relative to not retrieving at all, an important qualification to the general claim that grounding improves accuracy: grounding improves accuracy *on average* and *conditional on adequate retrieval quality*, not unconditionally.

3.2 Synthesizing the Comparative Accuracy Picture

Table 1 summarizes the qualitative pattern that emerges across the reviewed studies. Two consistent findings recur across domains and studies. First, every study that directly compares grounded and ungrounded systems on a matched query set reports a substantial reduction in error rate for the grounded condition (12; 2 vs. 8; 11). Second, none of these studies report error elimination; residual error in the grounded condition ranges from roughly 15% to over 30% depending on domain, query difficulty, and system, indicating that grounding shifts the accuracy distribution rather than truncating it at zero.

3.3 Domain Sensitivity

The magnitude of the grounding benefit appears to interact with domain structure. Legal research queries are unusually well suited to grounding because the ground truth (statutory text, case holdings) exists as discrete, retrievable, authoritative documents with stable identifiers (citations), making retrieval targets well defined [2, 8]. Clinical knowledge queries are structurally different: much clinical reasoning requires synthesizing across guidelines, patient-specific context, and evolving evidence rather than retrieving a single authoritative passage, and Singhal et al. [13] show that even scaled, instruction-tuned parametric models leave meaningful comprehension and reasoning gaps versus clinician performance, suggesting that grounding in medicine must be paired with structured reasoning support rather than treated as a standalone fix. This domain sensitivity is an important qualification against over-generalizing legal-domain grounding results to

Dimension	Ungrounded LLM	Grounded (RAG) LLM
Knowledge source	Static parametric weights	Parametric weights + retrieved external evidence
Dominant error type	Extrinsic fabrication (invented facts, citations) [2, 4]	Intrinsic misattribution/misapplication of real evidence [4, 8]
Long-tail query performance	Degrades sharply with entity/topic rarity [9]	Comparatively stable across rarity [9]
Provenance	None; cannot cite verifiable source	Partial; can cite retrieved passage, but citation may still misstate content
Reported error range (legal domain)	58–88% hallucination on verifiable case queries [2]	17–33% hallucination in deployed RAG tools [8]
Sensitivity to system quality	Sensitive mainly to model scale/training data [13]	Sensitive to retriever quality, chunking, and generator faithfulness [1, 3]

Table 1: Qualitative comparison of accuracy-relevant properties of ungrounded versus grounded LLMs on domain-specific research queries, synthesized from the cited empirical studies.

all research-query settings.

4 Current Approaches and Developments

Several architectural developments aim to close the residual gap identified above.

Self-reflective and adaptive retrieval. Self-RAG [1] trains the generator to emit reflection tokens indicating whether retrieval is needed, whether a retrieved passage is relevant, and whether the generated segment is supported by the passage, allowing the system to skip unnecessary retrieval and to down-weight or discard irrelevant evidence rather than being forced to condition on it.

Advanced and modular RAG pipelines. Gao et al. [3] document a shift from "naive" RAG (single-pass retrieve-then-read) toward advanced pipelines incorporating query rewriting and decomposition prior to retrieval, re-ranking and filtering of retrieved passages, and iterative or multi-hop retrieval for questions requiring synthesis across multiple documents–all intended to improve the quality of the evidence the generator ultimately conditions on, and thereby to narrow the intrinsic-faithfulness gap documented in legal audits.

Atomic and provenance-aware evaluation. FActScore [11] operationalizes factual precision at the level of individual atomic claims verified against an explicit knowledge source, moving evaluation away from coarse binary hallucination judgments toward graded, source-attributable scoring, a methodological shift with direct relevance to domain-specific research settings where partial correctness (a real citation used to support a subtly wrong proposition) is common and under-captured by binary metrics.

Preregistered, product-level auditing. Perhaps the most consequential development for this paper's topic is methodological rather than architectural: Magesh et al. [8] demonstrate the feasibility and necessity of preregistered, independent, product-level empirical audits of closed, commercially deployed grounded systems, rather than relying solely on vendor claims or academic benchmarks built on open corpora. This establishes a template that could be extended to grounded systems in medicine, scientific literature review, and other high-stakes domains.

5 Research Challenges and Limitations

Several challenges qualify the strength of conclusions that can currently be drawn.

Benchmark-domain mismatch. Widely used hallucination and factuality benchmarks (TruthfulQA, HaluEval, FActScore) were constructed primarily around general or encyclopedic knowledge [6, 7, 11], and their results may not transfer cleanly to specialized professional corpora with different citation conventions, ambiguity structures, and stakes.

Closed commercial systems limit reproducibility. As Magesh et al. [8] note explicitly, the proprietary, closed nature of commercial grounded research tools makes systematic, independent verification of vendor accuracy claims difficult; findings from one audited product generation may not generalize to subsequent versions or competing products, and academic researchers typically cannot inspect the retriever, index, or prompting strategy underlying a deployed system.

Confounding of grounding with other factors. Comparisons between "grounded" and "ungrounded" systems in the wild frequently confound retrieval augmentation with other differences–larger or differently fine-tuned base models, different prompting strategies, or additional guardrail layers–making it difficult to isolate the causal contribution of grounding itself, as opposed to co-occurring engineering improvements [3].

Self-assessment unreliability. Models are generally poor at recognizing their own hallucinations even when grounded evidence is available in context [2, 6], undermining approaches that rely on model self-critique alone (including some implementations of self-reflective RAG) without independent, external verification.

Binary versus graded error measurement. Much hallucination reporting (including the widely cited legal-domain percentages) reduces complex responses to a binary hallucinated/not-hallucinated judgment, which can obscure whether an error is a minor misattribution or a consequential factual reversal; FActScore-style atomic scoring [11] partially addresses this but is not yet standard in domain-specific auditing.

Static snapshot evaluation. Because both retrieval indices and underlying LLMs are updated frequently, empirical hallucination rates measured at one point in time (e.g., a specific vendor tool version) may not reflect current performance, limiting the durability of any single study's quantitative conclusions, including the estimates reported in this paper.

6 Research Gaps and Future Directions

Building on the challenges above, we identify several priority directions for future research.

Cross-domain replication of legal-style audits. The preregistered, product-level auditing methodology of Magesh et al. [8] should be extended to grounded systems in medicine, scientific research assistance, financial analysis, and engineering, domains where Singhal et al. [13] and others suggest the grounding-accuracy relationship may differ structurally from law.

Standardized, provenance-aware accuracy metrics for domain queries. There is a need for benchmarks that combine the atomic-fact precision philosophy of FActScore [11] with domain-authoritative knowledge sources (statute and case databases, curated clinical guideline repositories, peer-reviewed literature indices) rather than general web or encyclopedic corpora.

Disentangling retriever quality from generator faithfulness. Future controlled studies should independently manipulate retriever recall/precision and generator faithfulness training (e.g., Self-RAG-style critique) to quantify their separate marginal contributions to end-to-end accuracy, building on the taxonomy proposed by Gao et al. [3].

Longitudinal and versioned evaluation. Given the rapid iteration of both base models and commercial retrieval products, the field would benefit from longitudinal audit protocols that track hallucination and accuracy rates across product versions over time, rather than single-snapshot studies.

User-facing calibration and abstention. Since neither grounded nor ungrounded systems reliably recognize their own errors [2, 6], research is needed on interface-level mechanisms–confidence calibration, mandatory citation display, or structured abstention–that help domain professionals appropriately discount model output rather than treating fluent, grounded-looking answers as verified.

Task-conditioned accuracy definitions. Accuracy for a domain research query is not a single scalar; future work should distinguish citation-existence accuracy, holding/finding-characterization accuracy, and applicability-to-facts accuracy as separate, jointly reported dimensions, particularly in law and medicine where each dimension carries distinct professional risk.

7 Conclusion

The comparative evidence reviewed here supports a qualified conclusion: grounding a large language model in external retrieval measurably and substantially improves response accuracy on domain-specific research queries relative to purely parametric, ungrounded generation, particularly for long-tail, low-frequency facts characteristic of specialized professional inquiry [5, 9]. The clearest available field evidence, drawn from the legal domain, shows a reduction in hallucination rates from a range of roughly 58–88% for ungrounded general-purpose chatbots to roughly 17–33% for deployed, RAG-based commercial legal research tools [2, 8]. However, this same evidence decisively rejects the stronger marketing claim that retrieval augmentation renders LLM output "hallucination-free." Grounding changes the character of error–from wholesale fabrication toward misattribution and misapplication of genuine retrieved evidence–more than it eliminates error outright, a pattern consistent with the intrinsic/extrinsic hallucination taxonomy of Ji et al. [4] and the atomic factual-precision findings of Min et al. [11]. Architectural developments such as Self-RAG [1] and advanced modular retrieval pipelines [3] offer promising, partial mitigations by making retrieval adaptive and evidence-conditioning more disciplined, but none yet close the gap to reliable, verifiable accuracy at the level professional research contexts demand. For researchers, practitioners, and institutions deploying LLMs on domain-specific research tasks, the practical implication is that grounded systems should be treated as accuracy-improving but not accuracy-guaranteeing, warranting continued human verification, provenance display, and the kind of independent, preregistered auditing pioneered in the legal domain, extended systematically to medicine, science, and beyond.

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